CBIR Skin Anomaly Retrieval

https://www.dermengine.com/blog/artificial-intelligence-healthcare

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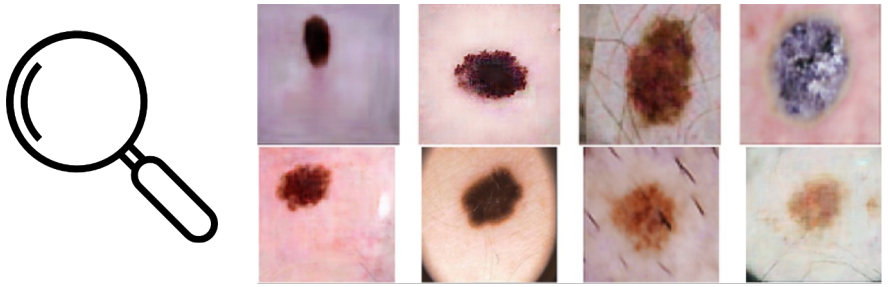
Motivation

Why Skin Anomaly Retrieval?

Dermatologists often compare lesions with previous cases
Large image datasets make manual search inefficient
Skin diseases show subtle visual differences



→ **CBIR can automatically retrieve visually similar cases**

Benefits

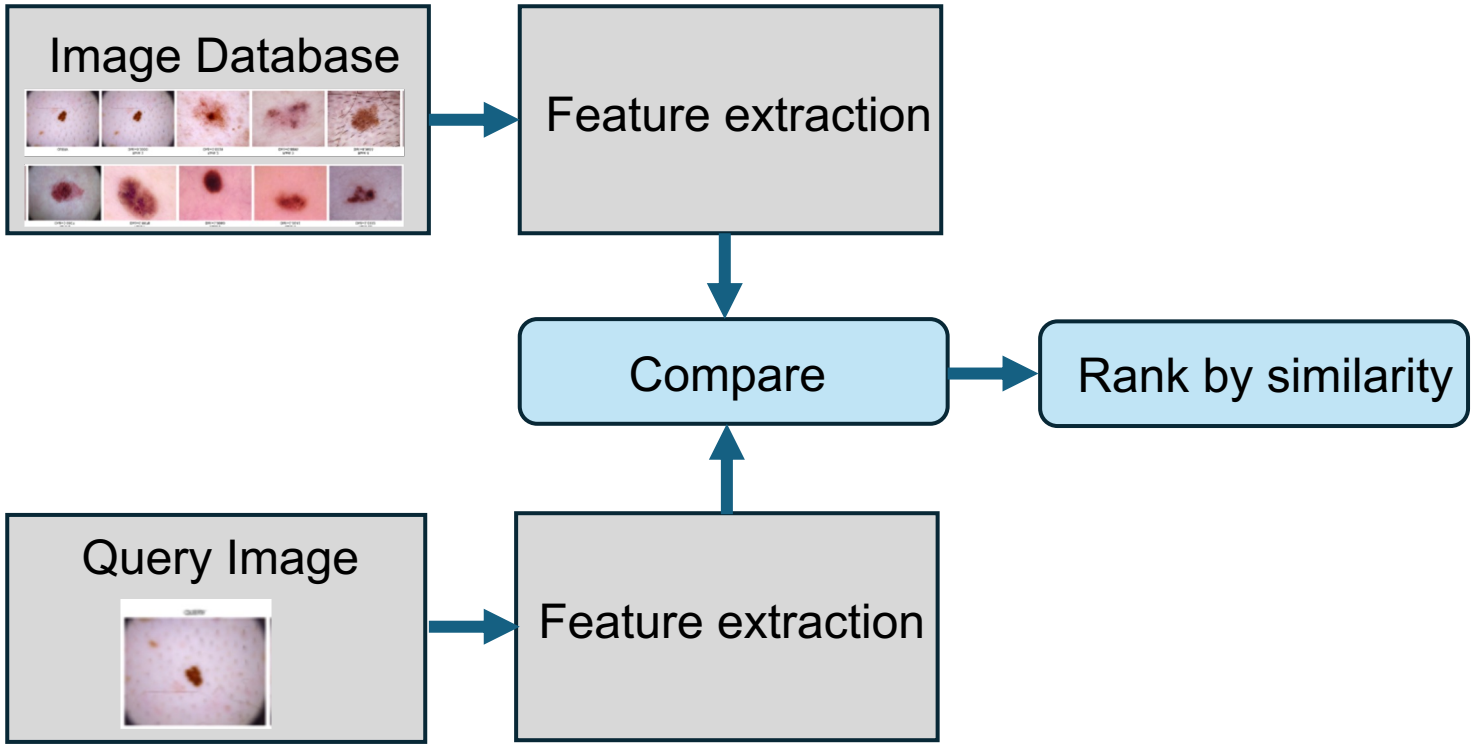
Assist dermatologists in diagnosis
Support early detection of malignant lesions
Enable efficient search in medical image databases
Useful for education and training

CBIR Concept

Content-Based Image Retrieval (CBIR)

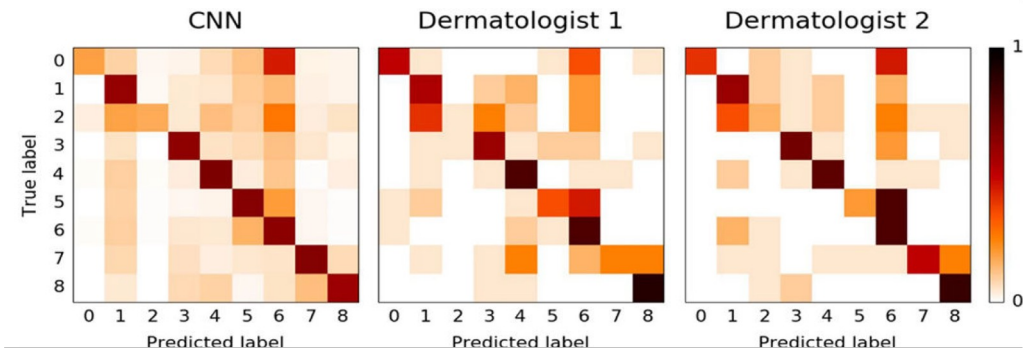
CBIR retrieves images based on **visual features**:

1) Input query image
2) Extract visual features
3) Compare features with database images
4) Rank images by similarity
5) Return most similar images



CBIR in Medical Imaging

Key research findings



Deep learning for skin lesion analysis

- CNNs can learn complex lesion patterns
- Dermatologist-level classification accuracy shown in research

CBIR using deep features

- Deep CNN embeddings enable retrieval of visually similar lesions
- Retrieval-based systems provide interpretable results

CBIR as decision support

- Similar image retrieval improves diagnostic interpretation
- Useful for education and clinical training

Research papers

Tschandl et al. (2018): Diagnostic accuracy of content-based dermatoscopic image retrieval with deep classification features

Barhoumi & Khelifa (2021): Skin lesion image retrieval using transfer learning-based approach for query-driven distance recommendation

Sadeghi et al. (2019) Using content-based image retrieval of dermoscopic images for interpretation and education: A pilot study

CBIR adaptation for Skin Images

Adapting the CBIR Baseline

The system is adapted for dermatology images.

1. CNN Model

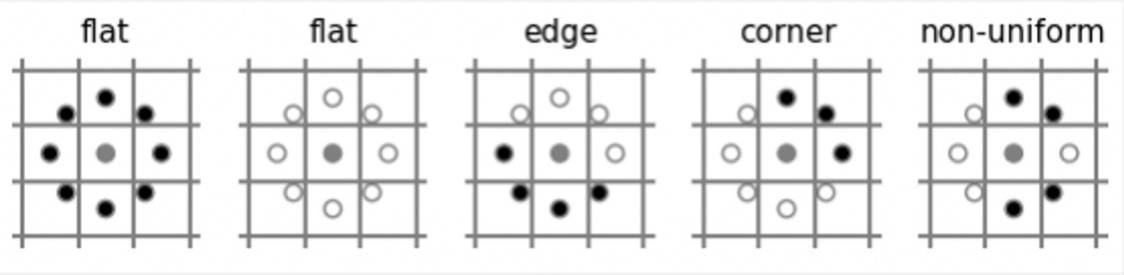
ResNet50 → EfficientNetB0
Better efficiency and strong feature representations

2. Texture Descriptor

Local Binary Patterns (LBP)

Captures micro-texture
Detects pigmentation irregularities
Suitable for skin lesion patterns

Local Binary Patterns (LBP)



Assess local texture around each pixel by comparing that pixel with its neighboring pixels

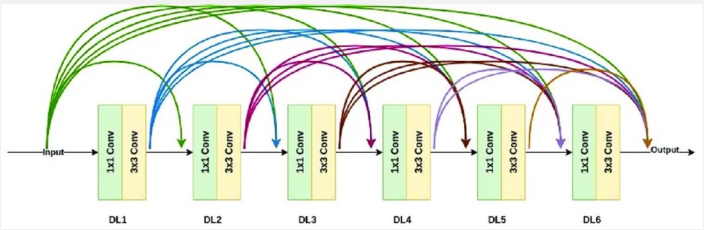
Feature Extraction

Image Descriptors:

The system extracts **three complementary features**:

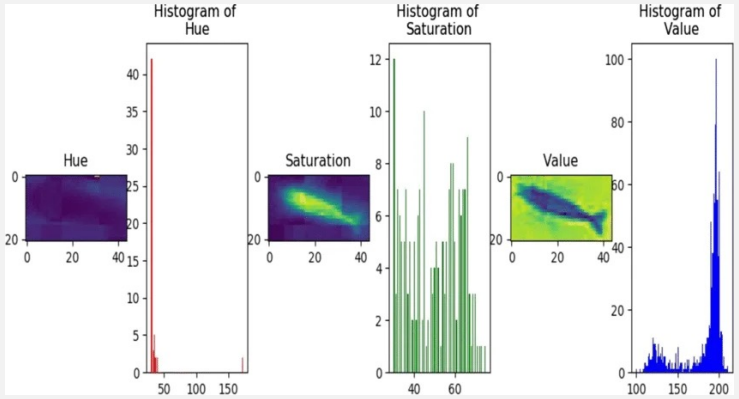
CNN Features

EfficientNetB0
Captures semantic patterns and lesion structure



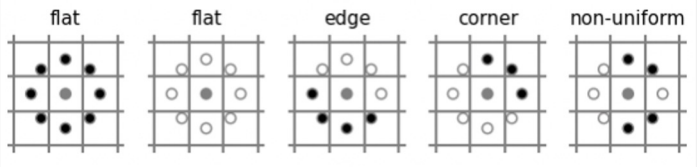
Color Histogram

HSV color space
32 bins per channel
Captures pigmentation and color distribution



Texture Descriptor (LBP)

Local Binary Patterns (LBP)
Captures micro-texture and surface irregularities



Feature fusion and indexing

Combining the multiple image descriptors

Early fusion
Normalised features are combined before similarity search.

Feature Concatenation
Feature vector: CNN features | Colour histogram | Texture descriptor

With feature weights (hyperparameters):

CNN features → **0.60**
Colour histogram → **0.25**
Texture descriptor → **0.15**

Similarity search
NMSLIB with HNSW indexing
HNSW is a graph-based nearest neighbour search algorithm

Feature	Captures
CNN features	high-level semantic patterns (lesion structure, shape)
Color histogram	pigmentation distribution
LBP texture	micro-texture and surface irregularities

↓

Normalisation

↓

Weighted concatenation

↓

Graph based Nearest neighbour search (HNSW), Similarity using Euclidean distance

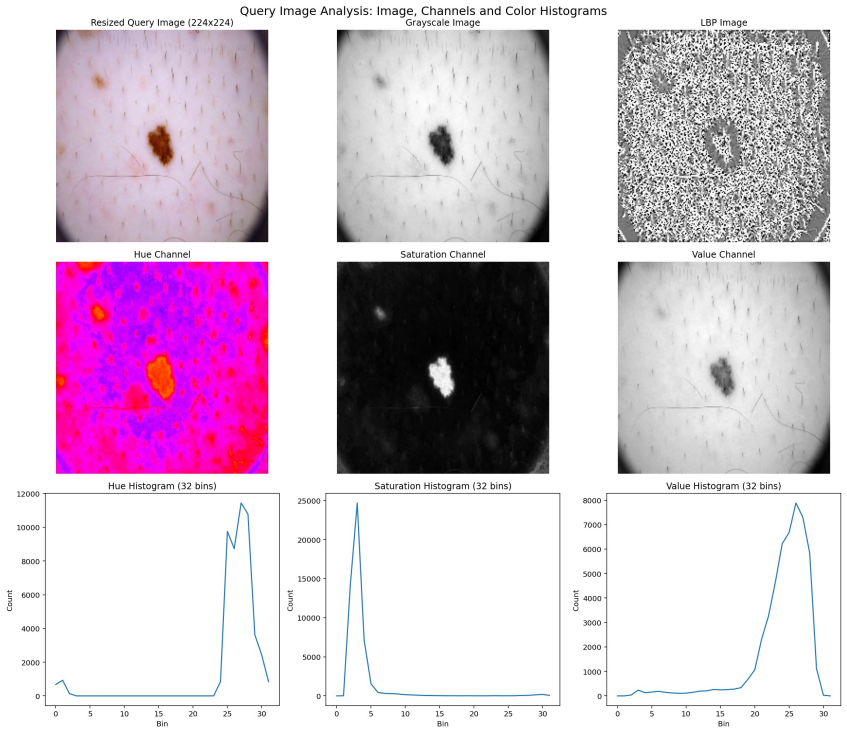
Results

Retrieval Output

- Query image
- Ranked list of most similar images
- Similarity distances

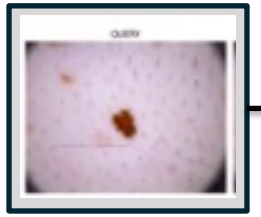
Example output:

Rank	L2 Distance	Image
1	0.3763	ISIC_0026784
2	0.3936	ISIC_0026753
3	0.4523	ISIC_0025299



Query image descriptors (CNN, HSV colour, LBP)

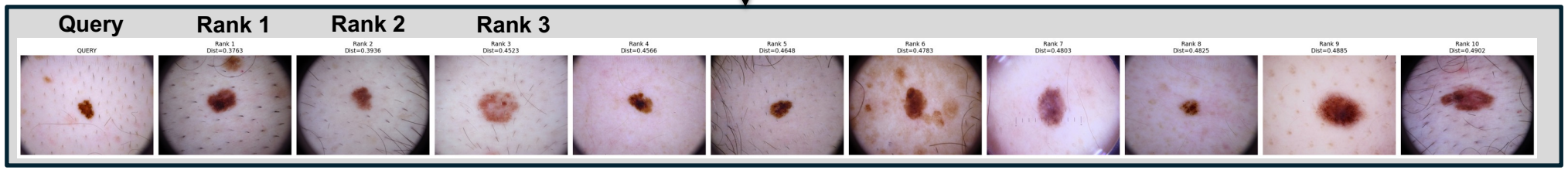
Query image



CBIR

Database (10.000 images): https://www.kaggle.com/code/jaimemorillo/cbir-skin-cancer-lesions/input

Ranked similar images



Conclusion

It is demonstrated how CBIR can be used for **skin anomaly retrieval**.

Key components:

- CNN features (EfficientNetB0)
- HSV colour histograms
- Local Binary Patterns (LBP) texture descriptors
- Weighted feature fusion
- HNSW indexing for fast retrieval (10.000 images)

Results

- Retrieval of similar skin anomalies assist dermatologists

Future Work

- CNN models trained specifically on skin lesions
- Evaluation using retrieval metrics (precision, recall)